TAMBAN THESIS - DEFENSE QUICK REFERENCE

Use this only as a memory aid. The full Defense Survival Pack is the source of truth.

Model 1	Find/localize parent fish. It does not decide quality.
Model 2	12 classes = A/B/C/Rejected × Body/Head/Tail.
Association	Attach each part to at most one fish. Ambiguous → UNASSIGNED.
Current weights	Body 50%, Head 30%, Tail 20% - still needs expert + ablation validation.
Missing evidence	Never renormalize it back to 100%.
Rejected	Positive unacceptable-quality evidence.
Ungraded	Insufficient/ambiguous evidence. Not a bad-quality class.
Current Model 2 lead	YOLO26m internal dev: mAP50-95 0.4606, mAP50 0.7876, F1 0.8310, 54.6 ms/img.
Do NOT call that	Final fish-level accuracy.
Biggest validation risks	Specimen leakage, B/C confounding, overlap ownership, background false positives, weight/gate justification.

Dangerous Questions - One-Line Answers

Question	Short defense answer
Why two models?	Model 1 establishes parent fish identity; Model 2 supplies grade-aware anatomical evidence.
Why 50/30/20?	It is a hypothesis being independently expert-elicited and validation-tested against alternatives.
What if parts overlap?	Ambiguous ownership can remain UNASSIGNED; do not force the nearest fish.
Tail not detected?	Not detected ≠ physically missing; treat as unavailable/unknown unless positive defect evidence exists.
Why Ungraded?	Abstention prevents weak or ambiguous evidence from becoming a fake confident grade.
mAP = accuracy?	No. mAP is detector performance; final fish-grade F1/accuracy is measured end-to-end.
Why YOLO over Faster R-CNN?	YOLO leads internal AP and latency; F1 is nearly tied; final selection still requires independent testing.

Question	Short defense answer
EfficientNet zero?	The tested configuration failed; we do not generalize that failure to the entire architecture family.
Why background errors?	Domain shift. We measure FP/image and use representative backgrounds/hard negatives rather than assuming robustness.
Final accuracy?	Only after the frozen pipeline runs once on an untouched specimen-independent final test set.

Never Say

- “Our model has 79% accuracy” when referring to mAP50.
- “50/30/20 is scientifically proven” before validation.
- “A missing Tail means Rejected.”
- “EfficientNet does not work.”
- “There is no leakage” without specimen-level proof.
- “The system works on any background.”